Career Launcher MBA

QA Numbers-03

Trailing Digits; Factorials

Question 1

The unit's digit of the final solution for $13 \times 27 \times 63 \times 51 \times 98 \times 46$ is

(1) 4

(2) 8

(3) 2

(4) None of the above

Question 2

The unit digit of y = (129 × 256 × 47$ × 1484) is 2.Then the digit in place of $ could be

(1) 7

(2) 3

(3) 5

(4) None of the above

Question 3

The unit's digit in the product of $(8267)^{153} \times (341)^{72}$ is

(1) 1

(2) 2

(3) 7

(4) 9

Question 4

Find the digit in the unit's place in the product 254 × 361 × 159 × 18.

(1) 1

(2) 6

(3) 4

(4) 8

Question 5

What is the unit's digit of $1^{781} + 2^{781} + 3^{781} + \ldots + 9^{781}$?

(1) 5

(2) 3

(3) 1

(4) 7

Question 6

The digit in the unit's place of the number represented by $(7^{95} - 3^{58})$ is

(1) 0

(2) 4

(3) 6

(4) 7

Question 7

What is the last digit of the number $3^{5^{7^{9}}}+1$?

(1) 1

(2) 7

(3) 4

(4) 3

Question 8

What will be the last digit of $2^{3^{4^5}} - 2^{3^{5^4}}$?

(1) 0

(2) 2

(3) 4

(4) 6

Question 9

What are the last two digits of $7^{(4004)}$?

(1) 21

(2) 61

(3) 01

(4) 41

Question 10

The unit's digit in $(743)^{85} – (525)^{37} + (987)^{96}$ is ____

(1) 9

(2) 3

(3) 1

(4) 5

Question 11

Let 20 × 21 × 22 × ..., 30 = A. If A is divisible by 10x find the maximum value of x.

(1) 3

(2) 4

(3) 5

(4) 6

Question 12

How many zeroes are there at the end of the following product?

$1 \times 5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50 \times 55 \times 60$

(1) 10

(2) 12

(3) 14

(4) 15

Question 13

If $15 \times 14 \times 13 \times ..... \times 3 \times 2 \times 1 = 3^m \times n$ where m and n are positive integers, then what is the maximum value of m ?

(1) 7

(2) 6

(3) 5

(4) 4

Question 14

How many trailing zeros would be there in 1024!

(1) 240

(2) 248

(3) 256

(4) 253

Question 15

If $a = 1^2$, $b = 2^3$, $c = 3^4$, ..., $z = (26)^{27}$ and P is the product of all the alphabets, how many zeroes exist at the end of P?

(1) 100

(2) 104

(3) 80

(4) 106

Question 16

If A is the product of first 100 multiples of 2 and B is the product of first 10 multiples of 5, then find the number of zeros at the end of A/B.

(1) 16

(2) 18

(3) 14

(4) 12

Question 17

How many consecutive zeros will be there at the end of the product $(2!)^{2!} \times (4!)^{4!} \times (6!)^{6!} \times (8!)^{8!} \times (10!)^{10!}$?

(1) 10! + 6!

(2) 2 (10!)

(3) 10! + 8! + 6!

(4) 6! + 8! + 2(10!)

Question 18

Consider the equation

$Y = (5!)^{5!} + (10!)^{10!} + (50!)^{50!} + (100!)^{100!}$

The number of zeros at the end of the whole expression would be _____

(1) 120

(2) 18000

(3) $(100!)^{100!}$

(4) None of the above

Question 19

For some natural number n, assume that (15,000)! is divisible by (n!)!. The largest possible value of n is

(1) 4

(2) 6

(3) 7

(4) 5

Question 20

The maximum value of the natural number n for which 21^n divides 50! is

(1) 6

(2) 7

(3) 8

(4) 9

Thanks!